

Question:

Most of the free fatty acids are transported in the blood:

Answer Choices:

- A. inside the red blood cells.
- B. combined with oxygen.
- C. as triglycerides.
- D. as free molecules.
- E. bound to white blood cells.
- F. as lipoproteins.
- G. attached to hemoglobin.
- H. combined with glucose.
- I. in the form of cholesterol.
- J. bound to albumin.

Reasoning + Answer:

</tool_call> reasoning process here

Free fatty acids in the blood are not transported as free molecules in large quantities, nor are they bound to red blood cells, white blood cells, hemoglobin, glucose, or cholesterol. While some fatty acids may be transported bound to albumin, the most accurate and physiologically relevant transport mechanism for free fatty acids in the bloodstream is through lipoproteins, such as chylomicrons and very-low-density lipoproteins (VLDL), which carry lipids including free fatty acids. Albumin can bind some free fatty acids, but the primary transport system for most free fatty acids is in lipoprotein particles. Therefore, the best choice among the given options is "as lipoproteins."

While albumin binding is a possible mechanism, the question specifies "most of the free fatty acids," and lipoproteins are the major carriers of free fatty acids in circulation.

<answer>F</answer>

